

Signal stability of high-Z sensors at synchrotron light sources

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ABSTRACT: The use of high-Z sensors for scientific applications at synchrotrons will gain more importance due to the upgrade of many light sources around the world to diffraction limited light sources for which the photon flux at high photon energies (above 30 keV) will significantly increase. Especially when illuminated with high photon fluxes, many established high-Z sensor materials like GaAs or Cd(Zn)Te are known to suffer from charge (de-)trapping effects in the bulk or at the contact interface which will ultimately lead to polarization and afterglow effects in the sensor. Commercially available high-Z sensors (Chromium-compensated GaAs, CdTe and CdZnTe) have been investigated at the Material Science beamline of the Swiss Light Source regarding their dynamic behavior when exposed to photon fluxes between 10^6 ph/mm²·s and $\approx 5 \cdot 10^{10}$ ph/mm²·s. In this work, the signal stability during irradiation will be presented, and a follow-up publication will discuss the afterglow.

KEYWORDS: Hybrid detectors; X-ray detectors

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1 Introduction

The most common sensor material for X-ray detectors at light sources is silicon due to its outstanding crystalline quality and charge transport properties. The biggest drawback for using silicon for higher photon energies is that its stopping power drops dramatically for photon energies above 15 keV due to its low atomic number of $Z_{Si} = 14$.

High-Z sensors, such as chromium compensated GaAs ($Z_{Ga/As} = 31/33$) or Cd(Zn)Te ($Z_{Cd/(Zn)/Te} = 48/(30)/52$), have much higher atomic numbers and therefore provide reasonably high stopping powers ($> 50\%$) for photon energies above 50 keV for typical sensor thicknesses [1]. However, high-Z sensors are compound semiconductors and suffer from other short-comings compared to silicon like a reduced uniformity, worse charge transport properties and especially charge (de-)trapping leading to afterglow and polarization effects [2–5]. Moreover, heavy materials produce relatively long ranged auto-fluorescence photons which can affect both the energy and spatial resolution of the detector.

Currently, many light sources around the world are being upgraded to diffraction limited light sources, increasing the monochromatic coherent photon flux for photon energies above 10 keV significantly [6]. This motivates the development of high-Z sensors for detectors at light sources to increase the absorption efficiency for photon energies above 15 keV. However, the signal quality may be reduced due to afterglow and polarization phenomena these sensors display which get more pronounced when they are exposed to high photon intensities [7–9].

The aim of the work described in this paper was to evaluate the dynamic behavior of various high-Z sensor materials to identify the most suitable alternative to silicon sensors for photon energies above 15 keV at synchrotron sources and FELs. The research goal was to investigate the dynamic behavior of the high-Z sensors which was separated into two phases: 1) the signal stability during the

irradiation phase to see, if e.g., a sensor suffers from polarization effects and 2) the signal transient after the irradiation phase, such as measuring the magnitude and duration of the afterglow from irradiation after closing the beam shutter.

2 Experimental setup

2.1 Sensors

Table 1 summarizes the sensors that have been investigated. Different variations of the high-Z sensors were used which will be explained in the following subsections. The employed silicon sensors were standard p-in-n sensors.

2.1.1 GaAs:Cr

Chromium compensated GaAs sensors were delivered by the Tomsk State University (TSU) in Russia. In order to obtain high-resistivity material, commercially available, low resistivity n-type GaAs wafer are annealed with chromium which diffuses into the GaAs wafer [10]. GaAs:Cr sensors work as photo resistors, meaning the dark current through the bulk is typically significantly higher compared to photo diodes. Extensive characterization results with GaAs:Cr sensors have previously been published [11, 12]. One of the main findings was the short lifetime of holes in the order of magnitude of a few ns.

For this study, different types of sensors have been tested, which include variations of the production batch as well as the underlying growth technique of the commercially available n-type GaAs wafer. Sensors from the production batches from 2017, 2018 and 2019 were produced from LEC-grown (*Liquid Encapsulated Czochralski*) GaAs wafers. VGF-grown (*Vertical Gradient Freeze*) GaAs wafers were used for the 2021 production batch. The thickness of the sensors was 500 μm and the typical sensor bias voltage is -300 V .

2.1.2 CdTe

The CdTe sensors investigated in this study came from Acrorad [13]. Acrorad offers two contact configurations, which allow the operation of the sensors either as photo resistor (quasi-ohmic type) or as photo diode (Schottky type). In case of the Schottky type CdTe sensors, the Schottky contact, consisting of Al, was on the pixel side in order to be able to collect electrons. The backside contact was made of Pt. Two batches of Schottky sensors were investigated from 2017 and 2019. The quasi-ohmic type CdTe sensors have Pt contacts on both sides. All CdTe-based sensors are 750 μm thick and were operated with -500 V .

2.1.3 CdZnTe

The CdZnTe sensors were obtained from Redlen Technologies [14]. Two different kinds of sensors are available, called ‘spectroscopic type’ and ‘high flux type’, the latter is optimized to handle higher photon fluxes compared to the ‘spectroscopic type’ by better matching the $\mu \times \tau$ products of electrons and holes [15, 16]. The thickness was 1500 μm and the operating sensor high voltage -500 V .

Table 1. Overview of the sensor types used for investigating their dynamic behavior.

Sensor material	Supplier	Thickness	Typical sensor high voltage	Variations
Silicon	Hamamatsu	320 μm	+200 V	- p-in-n
GaAs:Cr	Tomsk State University (TSU)	500 μm	−300 V	- LEC n-type wafer as base material in sensor batches from 2017/2018/2019 - VGF n-type wafer as base material in sensor batch from 2021
CdTe	ACRORAD	750 μm	−500 V	- Schottky type (electron collecting), two sensor batches 2017/2019 - Ohmic type
CdZnTe	Redlen	1500 μm	−500 V	- Spectroscopic type - High flux type

2.2 Readout chip

The readout chip, that was used for the experiments, was the JUNGFR AU ASIC. JUNGFR AU consists of 256×256 pixels with a pixel pitch of $75 \times 75 \mu\text{m}^2$ [17]. It was designed having in mind the high intensities exhibited at free electron lasers, thus being able to automatically adjust the gain of the preamplifier to the number of incoming photons on a pixel level by means of three different gain ranges. This mode, called ‘dynamic gain switching’ mode, was initially not foreseen for electron collection, and hence high-Z sensors, such that dynamic gain switching in the earlier generation of JUNGFR AU only works for hole collection. At the time of writing this article, two different iterations of JUNGFR AU existed, JF1.0 and JF1.1, with the most relevant difference being the ability of JF1.1 to automatically adjust the gain also in case of electron collection. For the experiments presented in this article, both iterations of the readout chips were used.

Alternatively, it is also possible to globally predefine and fix the gain range before each measurement which was the most common operating mode used in this study. In contrast to dynamic gain switching mode, where the CDS buffer (=Correlated Double Sampling) is only used for G0, the CDS buffer is always active in fixed gain mode having the advantage of an improved noise performance of the gain ranges G1 and G2 at the cost of a reduced dynamic range.

In order to be able to simultaneously use hole-collecting sensors (like Silicon) and electron-collecting sensors (like high-Z sensors), the working point has been chosen to be in the middle of the dynamic range, thereby sacrificing the maximum dynamic range. For the fixed gain mode, the values for the maximum linear dynamic range and noise performance measured with a GaAs:Cr sensor are given in table 2.

When using 20 keV photons and the typical integration time of 5 μs used for this study, the maximum linear dynamic range in fixed G2 of 25000 keV corresponds to a photon flux of $4.4 \cdot 10^{10} \times 20 \text{ keV}$ photons per mm^2 and second. If the pixel is operated in dynamic gain switching mode and the working point is optimized to collect only one kind of charge carrier, this maximum linear dynamic range will increase by roughly a factor of 4. (However, it is possible to measure higher photon fluxes, thereby sacrificing linearity of the readout electronics.)

Table 2. Overview of the maximum dynamic range and noise performance for different integration times when using fixed gain mode using JUNGFRAU1.0 in combination with a GaAs:Cr sensor ($U_{\text{Sensor}} = -300 \text{ V}$, $T = +15^\circ\text{C}$) as in this work.

Fixed Gain	Dynamic range [in keV]	Noise performance [ENC in keV, rms]
HGO (Super High Gain)	50 keV	$< 0.39 \text{ keV}$ ($5 \mu\text{s}$)
GO (High Gain)	120 keV	$< 0.47 \text{ keV}$ ($5 \mu\text{s}$)
G1 (Medium Gain)	2000 keV	$< 3.9 \text{ keV}$ ($5 \mu\text{s}$) $< 5.2 \text{ keV}$ ($300 \mu\text{s}$)
G2 (Low Gain)	25000 keV	$< 49.5 \text{ keV}$ ($5 \mu\text{s}$) $< 50.5 \text{ keV}$ ($300 \mu\text{s}$)

2.3 Test systems

The test systems consist of single chip assemblies, which refers to a single readout chip sized sensor of $2 \times 2 \text{ cm}^2$ bump bonded to a single readout chip. In case of silicon and GaAs:Cr sensors, the bump bonding has been done in-house at PSI, for all other sensors this has been done at ADVAFAB Oy [18]. As shown in figure 1, the testbed for the experiments was a standard JUNGFRAU module, which was modified to house three single chip assemblies instead of a full size ($4 \times 8 \text{ cm}^2$) silicon sensor. Each sensor can be individually supplied with high voltage. More details can be found in [12, 19].

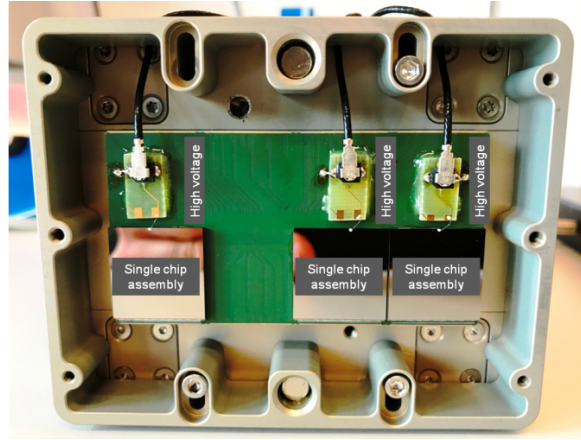


Figure 1. Testbed consisting of three independent single chip assemblies. The test system is derived from a standard JUNGFRAU multi-ASIC module.

2.4 Experimental setup and data treatment

The measurements were performed at the Material Science (MS) beamline of the SLS [20]. Due to constraints of the beamline, the photon energy for all measurements was 20 keV to ensure a maximum monochromatic photon flux of around $1 \times 10^{11} \text{ ph/mm}^2 \cdot \text{s}$. The photon flux was adapted to the needs of the experiment by means of different filters (mostly silicon filters of different thicknesses) to obtain intensities in a range between $1 \times 10^6 \text{ ph/mm}^2 \cdot \text{s}$ and around $5 \times 10^{10} \text{ ph/mm}^2 \cdot \text{s}$. For each measurement

(signal stability and afterglow), eight different intensities were used and, in order not to interfere with the previous irradiation, each irradiation was performed on a previously unirradiated section of the sensor. That results in a total of 16 regions of interest ($8 \times$ signal stability, $8 \times$ afterglow). The dimension of a region of interest (=ROI) was defined by slits and the size was typically around $1.2 \times 2.0 \text{ mm}^2$, corresponding to roughly 400 pixels ensuring reasonable statistics for each irradiation region.

Each sensor was measured with the same experimental procedure: initially the afterglow was probed by illuminating the sensors with three different exposure times (1 sec, 10 sec and 60 sec), followed by the signal stability test. This procedure was then repeated for three different temperatures of 0°C , $+15^\circ\text{C}$ and $+30^\circ\text{C}$ (see figure 2):

- *Afterglow measurements:* one irradiation cycle consists of 8 different photon fluxes on 8 different ROIs and a fixed irradiation time. The duration of the irradiation phase was 1 second in the first cycle, 10 seconds in the second cycle and 60 seconds in the third cycle. After each irradiation, the beam shutter was closed and the signal was measured for at least 100 seconds, before moving to the next ROI. The frame rate was 500 Hz, i.e., the acquisition period was 2 ms.
- *Signal stability test:* using the same 8 intensities that were used in the afterglow test, 8 new ROIs were irradiated for 20 minutes, measuring the signal with a frame rate of 200 Hz. In few cases, the maximum irradiation time was reduced due to time limitations.

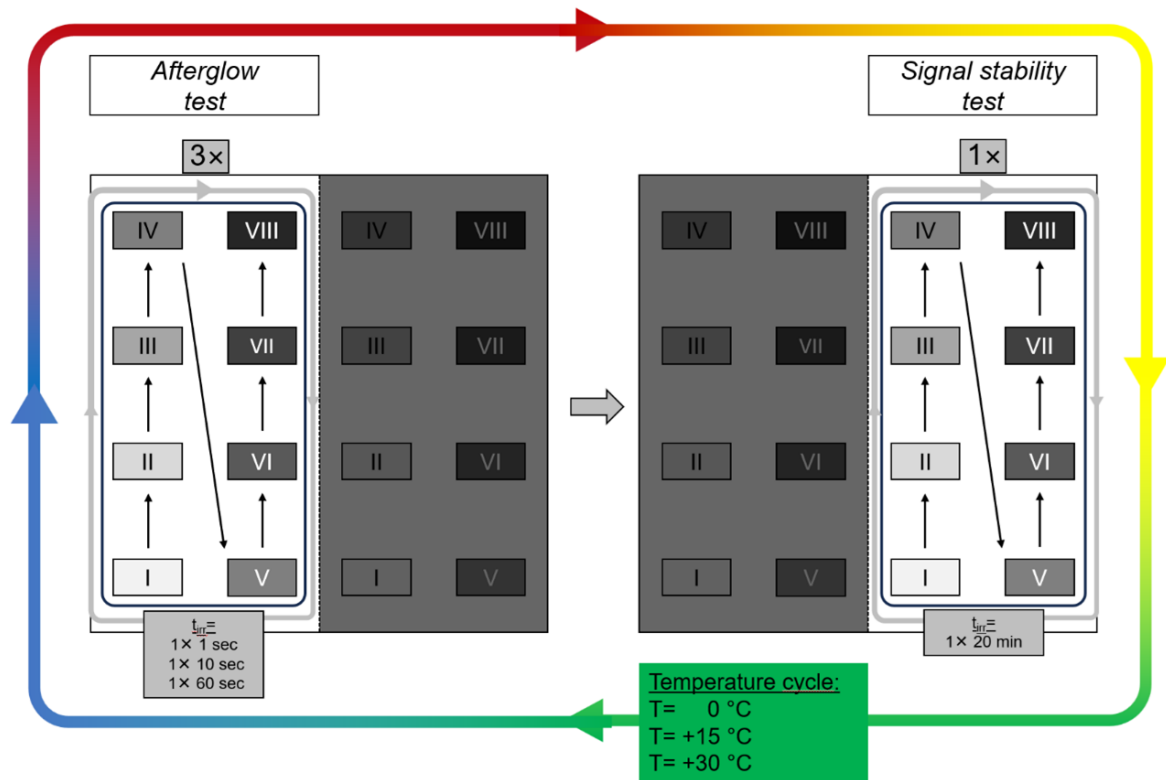


Figure 2. Schematics of the dynamics test. On the left half of a sensor, three cycles of afterglow measurements using different irradiation times ($= t_{\text{irr}}$) of 1 second, 10 seconds and 60 seconds, followed by a period with closed beam shutter of at least 100 sec were done. On the right half of the sensor, the signal stability test was performed. Each ROI was irradiated for (up to) 20 minutes.

As mentioned earlier, this manuscript will only report on the results of the signal stability test. The results of the afterglow measurements will be published in a follow-up work.

The analog signal output of the JUNGFR AU readout chip is digitized by means of an analog-digital-converter (ADC), providing a digital value in the unit of an analog-digital-unit (ADU). The raw data in units of ADU coming from the detector was corrected in the following order:

- *Pedestal correction:* as JUNGFR AU is a charge-integrating detector, the raw dark signal is mainly defined by the working points of the various buffers and by the dark current being collected from the sensor which strongly depends on the sensor material. The dark signal is called pedestal and ideally a constant value. The pedestal is measured by acquiring (at least) 1000 dark frames. For each pixel, the values of the dark frames are put into a histogram and the resulting distribution fitted by a Gaussian function. The mean value of the fit function corresponds to the pedestal value for the respective pixel. The r.m.s. from the Gaussian fit is the noise value of the pixel. To perform a pedestal correction, the pedestal value is subtracted from the measured pixel values on a frame-by-frame basis.

A pixel was considered malfunctioning, if its pedestal value was out of a pre-defined range or if its noise value was above or below an acceptance window of $\pm 10 \times$ the r.m.s. of the global noise distribution. Malfunctioning pixels were excluded from the data analysis.

- *Common mode correction:* common mode is a form of noise that is causing a temporal change of the pedestal value affecting several pixels at once and which is often correctable by tracking the pedestal. In JF1.0, the most occurring common mode contribution is a global shift of the pedestal value due to a readout buffer which was improved in the subsequent versions of the readout chip. For each assembly under test, the global pedestal shift of each frame was determined by measuring the difference of the five outermost pixel rows and columns from the previously obtained average pedestal value. The obtained correction value was then subtracted from the pedestal corrected image for all pixels.
- *Gain correction:* the output of each frame is given in units of ADU. In order to convert the output into keV and to furthermore take into account variations of the gain of the different pixels, the gain of each pixel is measured by illuminating the sensor with monochromatic X-rays. The photo peak position of the pedestal corrected monochromatic spectra (in ADU) was fitted for each pixel. By knowing the photon energies, a pixelwise conversion map from ADU into keV, called gain map, can be calculated. The pedestal corrected values of a pixel in ADU, giving the net collected charge of each pixel without contributions from dark/leakage current, are divided by the gain value to obtain the measured photo signal in units of keV.

The previously described process of gain correction applies to pixels in high gain ‘G0’. In the dynamic gain switching case, for pixels in medium gain ‘G1’ or low gain ‘G2’, the gain value of each pixel was calculated by taking the high gain value divided by the nominal gain ratios of $G0/G1 = 28.6$ (gain ratio: $G0/G1$) and $G0/G2 = 364.0$ (gain ratio: $G0/G2$) [21]. In case of using the fixed gain mode, the aforementioned gain ratios need to be divided by the CDS gain of 1.66. A specific calibration of the medium and low gain has not been performed. As measurements have shown that the temperature has a negligible influence on the pixel gain, by default calibration datasets acquired at $+15^\circ\text{C}$ have been used.

Each frame was treated as mentioned above, providing frames with the net photo induced signal in units of keV. For each frame, the average signal of each ROI was calculated by summing up the signals of all pixels in the ROI and then dividing the value by the number of pixels.

A MYTHEN strip detector was used to independently monitor the intensity of the incoming photon flux to be able to separate effects of varying incoming beam intensity from fluctuations of the measured signal in the sensor e.g., due to polarization effects. The MYTHEN strip detector with silicon sensor was placed out of the direct beam to measure air-scattered photons. MYTHEN is a single photon counting strip detector. The counting threshold was set to 50 % of the incoming photon energy of 20 keV and the number of photons was continuously counted in acquisition windows of 10 seconds. As the MYTHEN detector sits outside of the beam, the number of counts obtained by the MYTHEN detector do not represent the absolute photon flux. However, the photon counts from MYTHEN provide two important pieces of information: first, the photon counts from MYTHEN were used to monitor the signal stability of the beam during the measurement (with a 10 second precision). Secondly, it provides a relative measurement of the eight different illumination settings by calculating the ratios of the photon counts between the different intensities. These ratios were then taken to obtain the absolute photon fluxes by using the two lowest intensities measured by the JUNGFRU module in the direct beam as absolute intensity measurement. The two lowest intensities were chosen such that the JUNGFRU detector under test could be operated in high gain (G0), thus being able to provide precise values of the incoming photon flux. Additionally, the lowest photon fluxes ranged in the order of 10^6 – 10^7 ph/mm²·s, where no polarization effects were expected. If a silicon sensor was available in the hybrid module, the information of the silicon sensor (in direct beam) was used for the flux determination, considering the reduced absorption efficiency of the silicon sensor.

3 Results

As explained in previous sections, for each sensor under test, eight different signal transients (with photon fluxes between $\sim 5 \times 10^6$ ph per mm²·s and $\sim 5 \times 10^{10}$ ph per mm²·s) have been recorded. Typically, the integration time of the JUNGFRU readout chip was set to 5.0 μ s. However, in case of sensors with high dark current like GaAs:Cr or Ohmic CdTe the integration time was reduced at +30°C to 2.0 μ s to retain enough dynamic range. It is worth noting that the integration times are relatively long compared to the charge collection times which are in the order of few tens of ns and the preamplifier settling time of 500 ns. Therefore, in case of a partial shielding of the electric field in the sensor due to polarization, which would yield in prolonged charge collection times, most of the charge carriers would still be collected. In case of a single photon counting detector, the number of counts would decrease as the signal induction of the charge carriers would be extended in time, causing longer pulses with a reduced signal height.

The signal stability of the measured signal can be plotted as a function of time after opening the beam shutter. The average measured signal in keV per pixel was converted into the respective photon flux per mm² and second (ph/(mm²·s)) for each exposure. As the detector was operated at a rate of 200 Hz, the temporal resolution was 5 ms. The conversion allows to compare the measured photon flux with the incoming photon flux (which is measured independently by a photon counting strip detector in scatter mode, compare section 3.4).

For each assembly under test, the dependence of measured signal as function of the incoming photon flux was plotted and fitted with a linear for times of 10 ms, 100 ms, 1 s, 10 s, 100 s and 1000 s

after opening of the beam shutter. Additionally, the robustness of the charge collection over the whole irradiation time was determined by subdividing the irradiation phases into 1-second-long intervals. The 5 ms long measurements within these 1-second-long intervals were averaged, then the dependence of measured signal to incoming photon flux fitted with a linear function. The linear fits have been applied up to a maximum incoming photon flux of around 1.0×10^{10} ph/(mm²·s), up to which the response of the detector is linear.

From these linear fits, two conclusions can be drawn: (1) the linear relationship remains stable over time, indicating that charge collection is robust for photon fluxes up to approximately 1.0×10^{10} ph/(mm²·s), and (2) signal loss occurs for photon fluxes exceeding 3.0×10^{10} ph/(mm²·s).

3.1 High-flux type CdZnTe sensors

High-flux type CdZnTe sensors from Redlen are optimized for high photon flux applications by better matching the charge transport properties of electrons and holes compared to the more traditional spectroscopic type CdZnTe sensors [15, 16]. Compared to spectroscopic CdZnTe, where the ratio between the electron and the hole $\mu \times \tau$ product is roughly 500, this ratio is reduced to 3.8 for high-flux type CdZnTe, which is mainly achieved by reducing the electron lifetime in favor of the hole lifetime which is increased.

High-flux type CdZnTe performs close to the ideal behavior: throughout the irradiation time of 20 minutes, the measured signals remain stable (figure 3) for incoming photon fluxes between 6.7×10^6 ph/(mm²·s) and 3.3×10^{10} ph/(mm²·s).

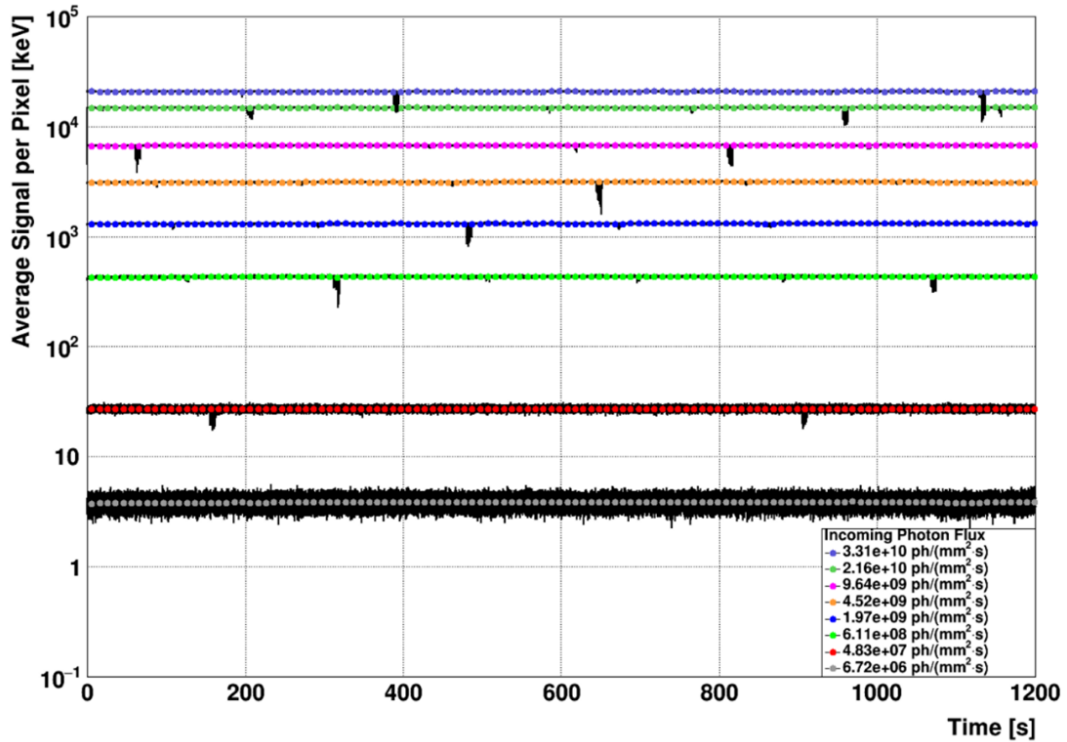


Figure 3. High flux type CdZnTe sensor (JF1.1 — Module-ID 343): average signal per pixel in keV of the different ROIs which have been illuminated with photon fluxes between 6.7×10^6 ph/(mm²·s) and 3.3×10^{10} ph/(mm²·s) at a temperature of $T = +15^\circ\text{C}$.

The dependence of measured photon flux as function of the incoming photon flux is linear, as depicted in figure 4 (left). The slight roll-off starting at photon fluxes above 4×10^{10} ph/(mm²·s) can be partly assigned to non-linearities as the dynamic range starts to saturate. For high-flux type CdZnTe, the measured signal drops to 94 % and 85 % (of the extrapolated signal) for incoming photon fluxes of 3.0×10^{10} ph/(mm²·s) and 5.0×10^{10} ph/(mm²·s), respectively, at temperatures of $T = +15^\circ\text{C}$. It seems that the signal loss is slightly more pronounced at higher temperatures (table 2). The margin of error for the extrapolated measured signal is estimated to be between 5–10 % due to the complex nature of the analysis, involving various cross calibration steps.

The slopes and offsets of the linear fits, which were extracted from the averaged signal of 1 second of illumination, are close to the ideal values ($m_{\text{ideal}} = 1$, $c_{\text{ideal}} = 0$) (figure 4, center/right). The slight deviations likely arise from uncertainties of the gain calibration. Remarkably, the fitting parameters show almost no dependence on temperature, suggesting that the charge collection in the sensor is highly effective and very little trapping/detrapping is taking place.

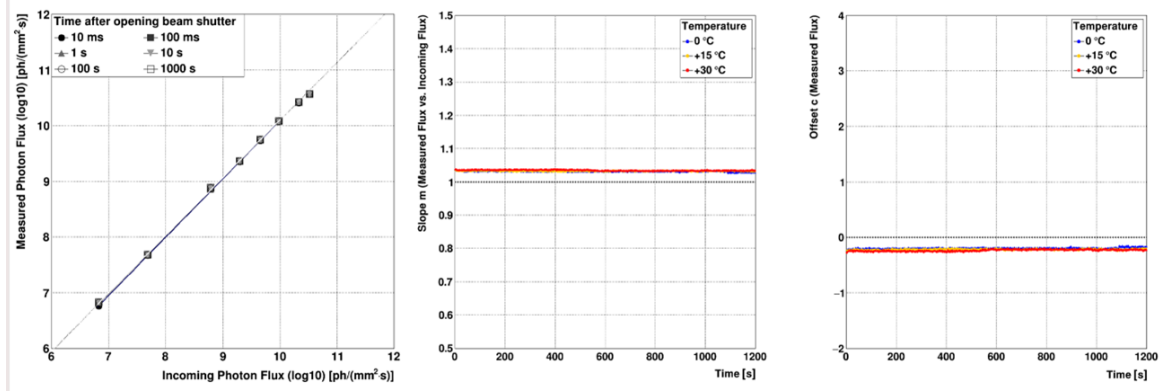


Figure 4. CdZnTe high flux type sensor ($T = +15^\circ\text{C}$) (left) Measured photon flux as a function of the incoming photon flux for different time stamps after opening of the beam shutter. (center/right) Linear fits of the dependence of the measured photon flux vs. incoming photon flux of the averaged values over 1 second of illumination. The slopes (m) and offsets (c) are constant over time as well as temperature. The deviation from the ideal values ($m_{\text{ideal}} = 1$, $c_{\text{ideal}} = 0$) is likely due to calibration uncertainties.

3.2 Spectroscopic type CdZnTe sensors

Spectroscopic type CdZnTe also shows promising characteristics regarding their signal stability, when irradiated with photon fluxes up to 4.3×10^{10} ph/(mm²·s). Similarly to high-flux CdZnTe, the transients indicate a robust measurement of the incoming photons without any signs of a collapse of the electric field (figure 5).

The incoming photons are reliably measured up to photon fluxes in the 10^{10} ph/(mm²·s) range, the extrapolated measured signal for photon fluxes above 1×10^{10} ph/(mm²·s) show similar values as for high-flux CdZnTe. For a photon flux of 3.0×10^{10} ph/(mm²·s) a measured signal of 102 % was estimated, which drops down to 86 % for photon fluxes around 5.0×10^{10} ph/(mm²·s) at $+15^\circ\text{C}$.

The fit parameters of the linear fit remain constant throughout the 20 minutes of irradiation with a slightly stronger temperature dependence compared to the high-flux CdZnTe.

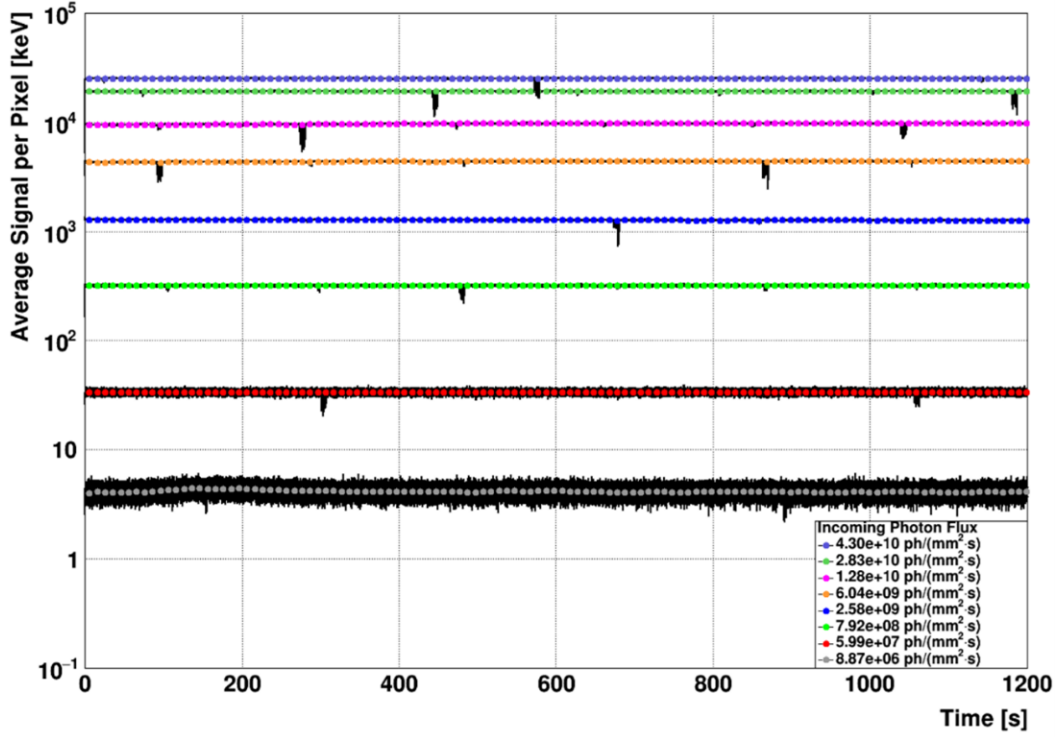


Figure 5. Spectroscopic type CdZnTe sensor (JF1.1 — Module-ID 401): average signal per pixel in keV of the different ROIs which have been illuminated with photon fluxes between $8.9 \times 10^6 \text{ ph}/(\text{mm}^2 \cdot \text{s})$ and $4.3 \times 10^{10} \text{ ph}/(\text{mm}^2 \cdot \text{s})$ at a temperature of $T = +15^\circ\text{C}$.

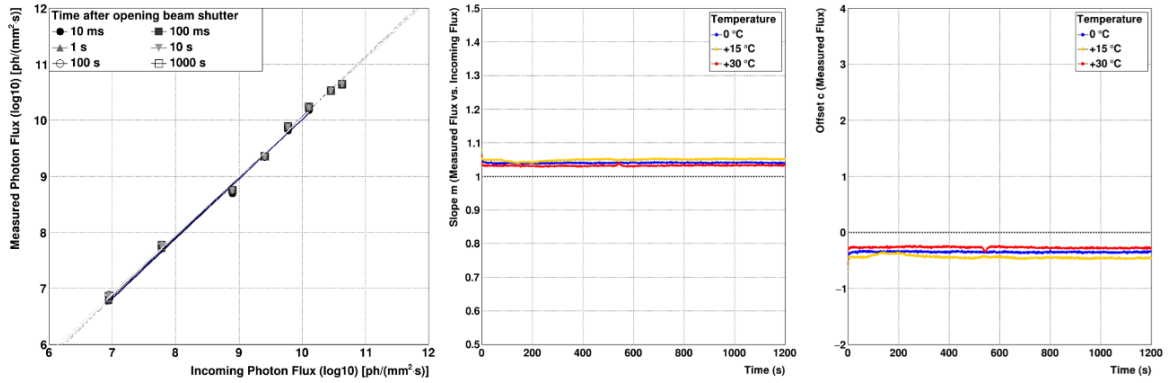


Figure 6. Spectroscopic type CdZnTe sensor ($T = +15^\circ\text{C}$) (left) Measured photon flux as a function of the incoming photon flux for different time stamps after opening of the beam shutter. (center/right) Linear fits of the dependence of the measured photon flux vs. incoming photon flux of the averaged values over 1 second of illumination. The slopes (m) and offsets (c) are constant over time as well as temperature. The deviation from the ideal values ($m_{\text{ideal}} = 1$, $c_{\text{ideal}} = 0$) is likely due to calibration uncertainties.

3.3 Chromium-compensated GaAs

Another promising, albeit with lower stopping power, high-Z sensor material for high-flux application is chromium-compensated GaAs, which has been intensively studied by the PSD Detector Group in the previous years [11, 12, 22].

Compared to CdZnTe, the charge transport properties of GaAs:Cr are inferior, especially the low hole $\mu \times \tau$ product which is two to three orders of magnitude less and in the order of $< 10^{-6} \text{ cm}^2/\text{V}$, causing negative signals in the vicinity of the collecting electrode due to incomplete positive charge collection. Another obvious disadvantage is the reduced stopping power of GaAs:Cr compared to Cd(Zn)Te due to its lower atomic number.

Currently, the biggest advantage of using GaAs:Cr is the commercial availability of 4-inch wafers allowing to produce full-scale sensors with a dimension of $8 \times 4 \text{ cm}^2$. Being the standard size of modules for all detectors produced by the PSD Detector Group, they can be easily integrated using existing mechanics. If the sensor quality permits, they could be used in areas with a high photon flux at moderately high photon energies (12–40 keV) instead of silicon sensors to prevent radiation damage to the readout chips.

During the measurement campaign, eight different GaAs:Cr sensors have been tested, six used LEC-grown GaAs wafers as base material, two were VGF-grown material. The general behavior of all GaAs:Cr sensors was similar; therefore this behavior will be shown exemplarily for a LEC-grown GaAs:Cr sensor from the 2018 production run, mounted on a JUNGFRAU1.1 readout chip (Module-ID 357). Figure 7 shows how the sensor handles continuous illumination of 20 keV photons in a range between $1.45 \times 10^7 \text{ ph}/(\text{mm}^2 \cdot \text{s})$ and $6.2 \times 10^{10} \text{ ph}/(\text{mm}^2 \cdot \text{s})$ at a temperature of $+15^\circ\text{C}$. There is no obvious drop of the measured signal which remains stable during the 20 minutes of irradiation.

The dependence of measured signal as function of the incoming photon flux is shown in figure 8, left. For photon fluxes up to $< 1 \times 10^{10} \text{ ph}/(\text{mm}^2 \cdot \text{s})$, the dependence is linear, while for higher photon fluxes, the dependence becomes sublinear. Remarkably, the reduction of measured signal appears to be an (quasi-) instantaneous effect, as no significant reduction over time of the measured signal for high photon fluxes is visible in figure 7. The general trend among all GaAs:Cr sensors

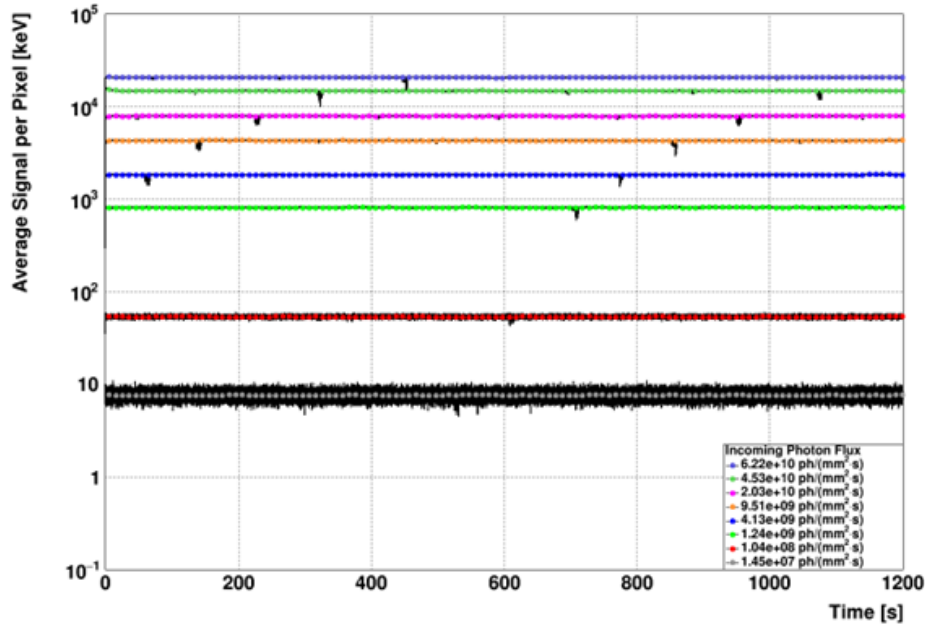


Figure 7. Chromium-compensated GaAs (2018) (JF1.1 — Module-ID 357): average signal per pixel in keV of the different ROIs which have been illuminated with photon fluxes between $1.45 \times 10^7 \text{ ph}/(\text{mm}^2 \cdot \text{s})$ and $6.2 \times 10^{10} \text{ ph}/(\text{mm}^2 \cdot \text{s})$ at a temperature of $T = +15^\circ\text{C}$.

indicates that the reduction of the measured signal is more pronounced compared to CdZnTe sensors (of both kinds). Moreover, there is a clearly visible difference of the linear fit parameters as function of the temperature, which was less apparent in case of CdZnTe (figure 8, center/right). It takes few seconds after the opening of the beam shutter at temperatures of 0°C and +15°C to reach their equilibrium state. It is worthwhile to note that the deviation from the equilibrium state is significantly more pronounced for the 0°C dataset compared to the +15°C dataset, while the +30°C is stable from the beginning of the measurement.

As the temperature dependence of the linear fit parameters is common for all GaAs:Cr sensors, but hardly present in CdZnTe, presumably the higher detrapping rate at elevated temperature is beneficial for the detector performance. However, due to the high dark current of GaAs:Cr sensors, operation at reduced temperatures is required, ideally below +15°C, for using them with charge-integrating readout schemes [12]. On the other hand, the higher dark current at elevated temperature will be easily tolerated by single photon counting readout chips, therefore higher sensor temperatures might be preferable for a stable detector performance.

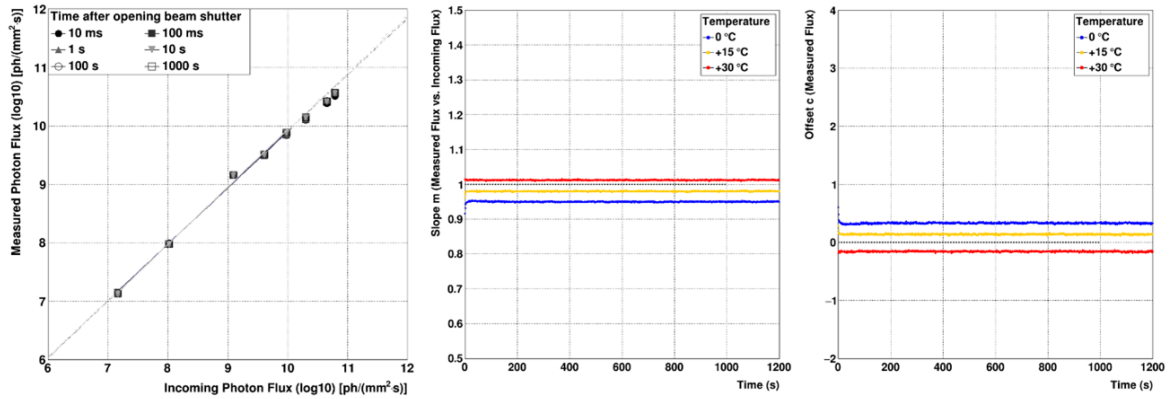


Figure 8. GaAs:Cr from 2018 (JF1.1 — Module-ID 357): (left) Measured photon flux as a function of the incoming photon flux for different time stamps after opening of the beam shutter. (center/right) Linear fits of the dependence of the measured photon flux vs. incoming photon flux of the averaged values over 1 second of illumination.

3.4 CdTe (electron collecting Schottky type)

Electron collecting Schottky type CdTe sensors were also tested, the results of one sensor are depicted in figure 9 as an example of the behaviour typical of these sensors. The transients over 1000 seconds of irradiation show that the measured signals strongly fluctuate over time which was a general observation in Schottky type CdTe sensors. However, in the log representation of the time axis (figure 9, right column), the signal in the first 100 ms after opening the beam shutter is stable and proportional to the incoming photon flux which might allow for measurements at pulsed sources like FELs.

In order to exclude a potential influence of bias-induced polarization effects, the measurements were performed with a continuously bias sensor (as it was the case for all other measurements) and with a refreshed sensor bias (30 seconds before each irradiation phase) [23]. However, the general observation of strongly fluctuating measured signals over time did not change.

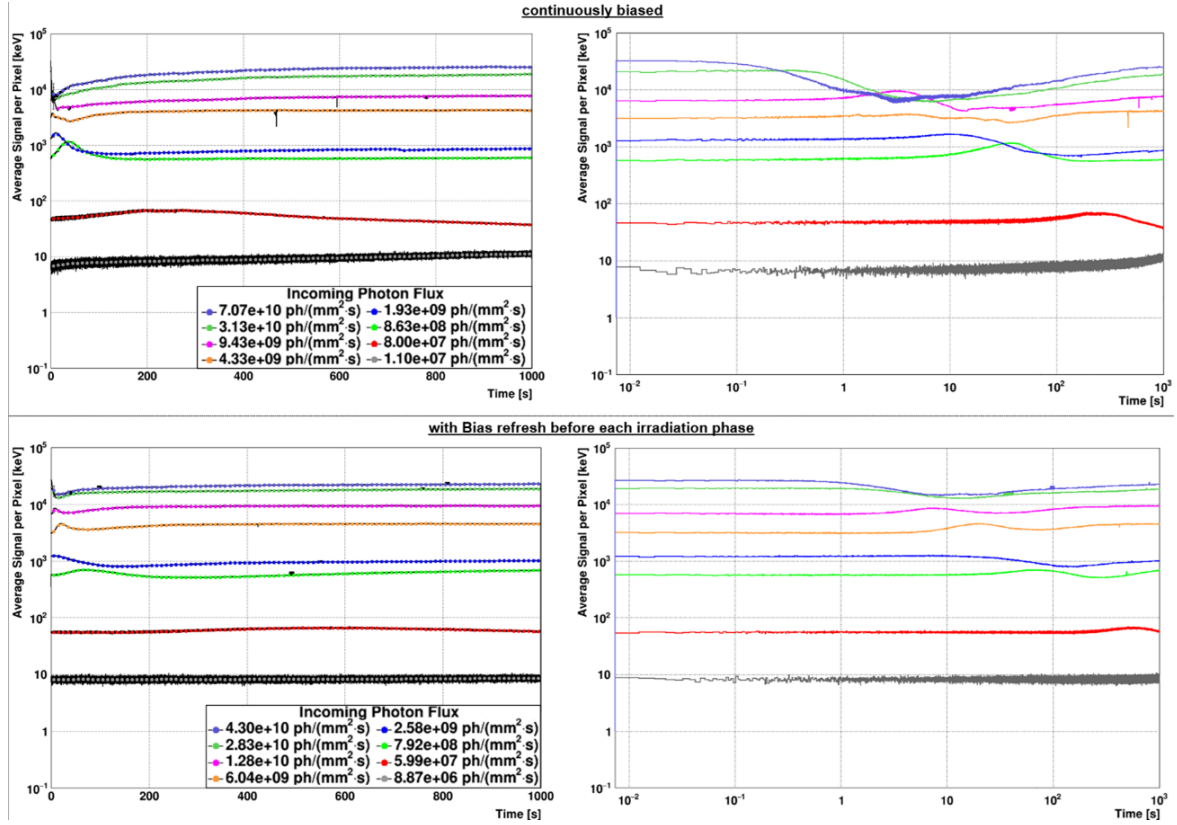


Figure 9. Electron collecting Schottky type CdTe (JF1.0 — Module-ID 295): transients of the average signal per pixel in keV of the different ROIs which have been illuminated with photon fluxes in the range between 10^7 ph/(mm²·s) and $\approx 5 \cdot 10^{10}$ ph/(mm²·s) at temperatures of $T = +15^\circ\text{C}$. In the top row, the sensor was continuously biased. In the bottom row, the bias has been refreshed before each irradiation phase to remove potential bias induced polarization effects which might occur in electron collection Schottky CdTe sensors.

3.5 CdTe (Quasi-ohmic type)

The transients of quasi-ohmic CdTe sensors exhibit a characteristic increase of the measured signal which starts to set in after few seconds of stable operation while continuously irradiated (figure 10). During irradiation, the signal never breaks down.

The fit parameters are closer to the ideal values ($m = 1$, $c = 0$) at the beginning of the irradiation, as especially the strong increase of measured signal at low intensities causes a significant deviation of the ideal linear response of the sensor. Moreover, a strong temperature dependence can be seen, it is unclear whether the fit parameters would converge to the same value after an extended period of time or whether a refresh of the sensor bias would have reverted the sensor performance to its behavior at the beginning of the irradiation.

4 Summary and conclusions

In conclusion, in none of the investigated regions of the various high-Z sensors the charge collection was severely hampered causing a significant break-down of the measured signal as it would be expected in case of a sensor polarization [8, 9].

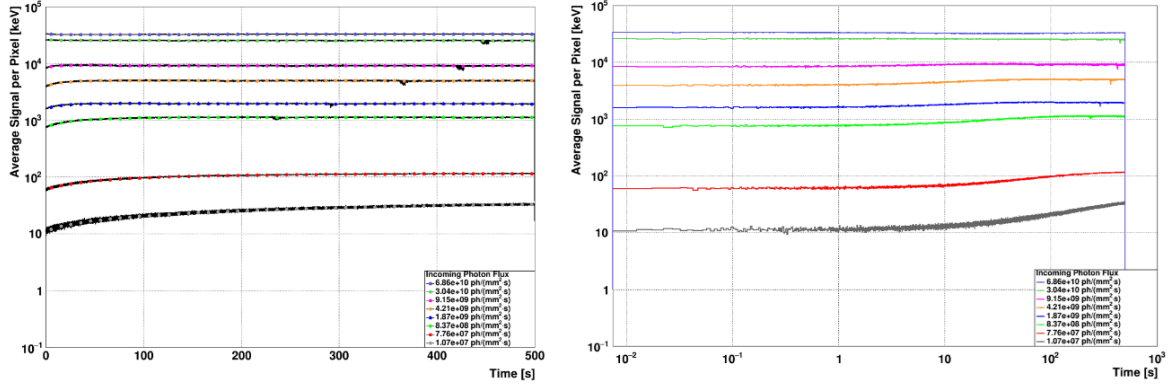


Figure 10. Quasi-ohmic type CdTe (JF1.0 — Module-ID 238): average signal per pixel in keV of the different ROIs which have been illuminated with photon fluxes between 1.1×10^7 ph/(mm²·s) and 6.9×10^{10} ph/(mm²·s) at a temperature of $T = +15^\circ\text{C}$.

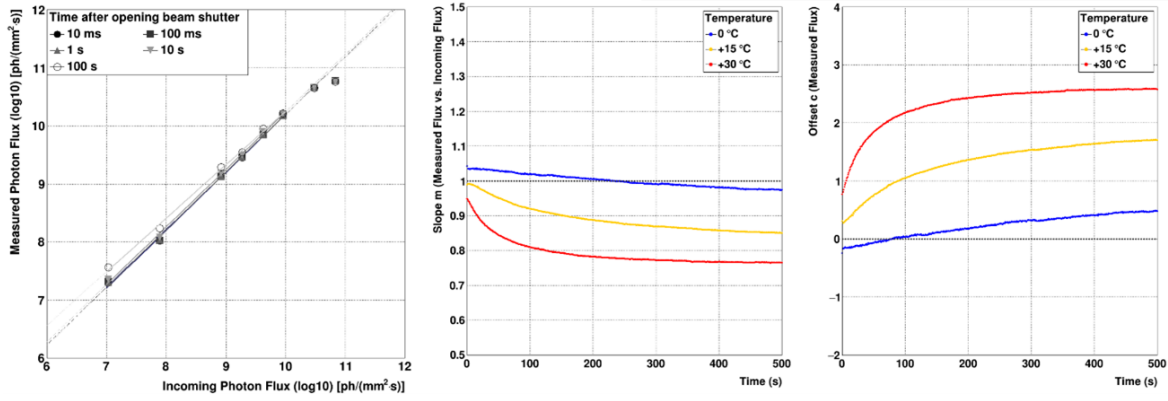


Figure 11. Quasi-ohmic type CdTe (JF1.0 — Module-ID 238): (left) Measured photon flux as a function of the incoming photon flux for different time stamps after opening of the beam shutter. (center/right) Linear fits of the dependence of the measured photon flux vs. incoming photon flux of the averaged values over 1 second of illumination.

However, CdTe sensors of either kind, electron collecting Schottky type and quasi-ohmic type, suffer from disturbed charge collection during phases of continuous illumination creating instable signal transients. Potentially, CdTe based sensors might be used in applications with short photon pulses, but this would require further tests at pulsed sources.

Both, chromium compensated GaAs and CdZnTe (high-flux type and spectroscopic-type), performed well during the high-flux illumination tests, while the latter performed better for photon fluxes above 3.0×10^{10} ph/(mm²·s) (see table 3). It is noteworthy that the signal loss for photon fluxes around 5×10^{10} ph/(mm²·s) is already close to the limit of the linear dynamic range and therefore part of the observed signal loss is due to a beginning saturation of the readout buffers.

The performance of the CdZnTe sensors regarding signal stability was very close to silicon sensors which are still considered the state of the art regarding their dynamic behavior. Remarkably, the 320–450 μm thick silicon sensors which were used for reference measurements suffered severely from the induced radiation damage during the irradiation tests, while the CdZnTe (but also GaAs:Cr) sensors performance was unaltered.

Table 3. Signal loss of measured signal from linearly extrapolated value for incoming photon fluxes of 3.0×10^{10} ph/(mm²·s) and 5.0×10^{10} ph/(mm²·s) (which is close to the limit of the linear dynamic range).

Module-ID	Sensor	Bias voltage	Measured signal (%) at 3.0×10^{10} ph/mm ² ·s			Measured signal (%) at 5.0×10^{10} ph/mm ² ·s		
			0°C	15°C	30°C	0°C	15°C	30°C
#016	GaAs:Cr (2016) - LEC	−300 V		89			84	
#077	GaAs:Cr (2017) - LEC	−300 V		45			35	
#238	GaAs:Cr (2017) - LEC	−300 V	79	72	66	63	61	66
#237	GaAs:Cr (2018) - LEC	−300 V	92	74	79	62	63	72
#357	GaAs:Cr (2018) - LEC	−300 V	91	76	84	79	69	72
#295	GaAs:Cr (2019) - LEC	−300 V	81	67	60	50	48	50
#528C4	GaAs:Cr(2021) - VGF	−300 V	99	87	104	87	77	80
#528C7	GaAs:Cr(2021) - VGF	−300 V	95	82	92	84	71	59
#528C7	GaAs:Cr(2021) - VGF	−450 V	93	108	115	83	89	93
#343	CdZnTe (High-flux)	−500 V	96	94	93	85	85	79
#401	CdZnTe (Spectroscopic)	−500 V	102	102	97	87	86	81

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